

Zstate Pilot Status Brief - July 2026 (WIP)

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Work in progress - 2026-07-03 git d55c5f1

Internal team share. Covers architecture, pilot tasks, reward design, PM simulator, and early model findings. Not for external publication.

1. What we are building

Zstate is an equity-research benchmark with two complementary tracks in one repo: Track A (public leaderboard, single-turn forensics/modeling tasks) and Track B (RL training env with a skeptical Portfolio Manager). Both share fixed EDGAR corpora, expert ground truth, and a fracture taxonomy.

Architecture (high level)

Shared foundation: Corpus (fixed doc bundles) + Expert pipeline (CFA draft/review) + Trajectory schema v1

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+-- Track A (benchmark_v0.1/) -- single-turn Type F/M tasks
|   -> 3-layer reward (L1 accuracy / L2 gold path / L3 citations)
|   -> Leaderboard + fracture codes
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+-- Track B (env_v1/) -- dual-control episodes
|   -> PM policy FSM (pushback, follow-ups, pushover trap)
|   -> 4-component reward (Outcome / Grounding / Defense / Hallucination)
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+-- Track C (future export/) -- curated JSONL + reward vectors for training product
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2. Reward systems

Track A - 3-layer benchmark score

Layer | What it measures | Typical weight (Type F)

L1 Hard accuracy | Python verify vs expert ground truth (metrics, arithmetic) | 55%

L2 Gold path | Section recall, tool order, workflow fidelity | 25%

L3 Trust / citations | Verbatim snippets, policy acks, anti-hallucination | 20%

Composite = weighted sum. Fracture codes tag failure modes (e.g. CITE_BROAD).

Track B - 4-component env reward

Reward = 0.45 Outcome + 0.25 Grounding + 0.20 Defense - 0.10 Hallucination. Defense is env-only: rewards substantive engagement when the PM pushes back. Capitulation without retrieval triggers pushover penalties.

PM simulator (Track B)

Scripted FSM (pm_v1_1): opening skepticism -> follow-up A/B/C branches -> pushover trap.

Agent tools: fixed corpus search + send_message_to_pm + submit_recommendation.

Sample episode: Solaris earnings-quality dispute (env_v1/episodes/).

Defense is the new discriminator vs single-turn QA - see frontier runs in env_v1/runs/.

3. Published pilot tasks (5/15 MVD)

Task ID | Ticker | Type | Tier | Objective

GOOGL_footnote_reconciliation | GOOGL | F | L2 | Using Alphabet Inc.'s Form 10-Q for the quarter ended March 31, 2026 (filed April 30, 2026), reconcile reportable seg...

PEP_fx_organic_growth | PEP | M | L3 | Using PepsiCo, Inc.'s Form 10-K for the fiscal year ended December 27, 2025 (filed February 3, 2026), compute constan...

AMZN_footnote_reconciliation | AMZN | F | L3 | Using Amazon.com, Inc.'s Form 10-K for the fiscal year ended December 31, 2025 (filed February 6, 2026), reconcile re...

NFLX_guidance_drift | NFLX | F | L3 | Using Netflix, Inc. materials for FY2025 guidance and Q3 2025 actuals, quantify guidance drift on **content cash

Task highlights

GOOGL (F): Q1 2026 segment sum + hedging reconciling item - ceiling task for frontier models.
PEP (M): FY2025 FX / organic growth decomposition from MD&A tables.
AMZN (F): FY2025 segment operating income bridge with stock-based comp decoy.
NFLX (F): FY2025 cash content guidance vs nine-month YTD actuals (guidance drift).
KO (F, new): FY2025 five-segment Total net revenues + Corporate + Eliminations bridge to \$47,941M consolidated (reconciliation_bridge_total, v0.2 naming); LatAm (2)% total vs (12)% FX from MD&A.

4. Eval methodology

Campaign runner: benchmark_v0.1/scripts/run_benchmark_campaign.py
Eval mode ON: generic citation rules only (no task-specific cheat-sheets).
Path roles: canonical slugs (segment_financials, narrative_fx) - not issuer note numbers.
Headline composite excludes GOOGL (ceiling); PEP + AMZN + NFLX (+ KO in 5-task campaigns).
Models routed: OpenAI (gpt-*), Anthropic (claude-*), Gemini (gemini-* via OpenAI-compatible API).
3 runs per task (median) for all models in pilot_eval_5task_v1.
v0.2 L3 anchors: ground-truth-derived row/column anchors with token-set matching; anchor regression + schema coherence gates in CI.
Synthetic L3 decoy diagnostic (Jul 3): 0 bait citations across 14 evaluated runs, 3 models - robustness checkmark, not yet a ranking dimension.

5. Findings (July 2026 pilot)

Campaign pilot_eval_5task_v1: 45 runs scored (5 tasks x 3 models x 3 runs). Score = median L1 hard-accuracy; Composite = median full L1+L2+L3 (3 runs per cell).

Task	Model	Score	Composite	Issue
GOOGL segment sum (ceiling) *	claude-sonnet-4-5	1.000	1.000	None
GOOGL segment sum (ceiling) *	gemini-2.5-flash	1.000	1.000	None
GOOGL segment sum (ceiling) *	gpt-4o	1.000	1.000	None
PEP FX / organic growth	claude-sonnet-4-5	1.000	1.000	None
PEP FX / organic growth	gemini-2.5-flash	1.000	1.000	Citation snippet not in bundle excerpt (CITE_HALLUC)
PEP FX / organic growth	gpt-4o	1.000	0.864	Citation snippet not in bundle excerpt (CITE_HALLUC); Non-auditable citation (CITE_BROAD)
AMZN segment bridge	claude-sonnet-4-5	1.000	1.000	None
AMZN segment bridge	gemini-2.5-flash	1.000	1.000	None
AMZN segment bridge	gpt-4o	1.000	0.958	Non-auditable citation (CITE_BROAD)
NFLX guidance drift	claude-sonnet-4-5	1.000	0.879	Citation snippet not in bundle excerpt (CITE_HALLUC); Non-auditable citation (CITE_BROAD)
NFLX guidance drift	gemini-2.5-flash	1.000	0.830	Non-auditable citation (CITE_BROAD); Citation snippet not in bundle excerpt (CITE_HALLUC); +1 more
NFLX guidance drift	gpt-4o	1.000	0.830	Citation snippet not in bundle excerpt (CITE_HALLUC); Non-auditable citation (CITE_BROAD)
KO segment bridge	claude-sonnet-4-5	1.000	0.896	Citation snippet not in bundle excerpt (CITE_HALLUC)
KO segment bridge	gemini-2.5-flash	1.000	0.873	Non-auditable citation (CITE_BROAD)
KO segment bridge	gpt-4o	1.000	0.873	Non-auditable citation (CITE_BROAD)

* Not in headline rank (see note below).
Why GOOGL is scored but excluded from headline rank: GOOGL_footnote_reconciliation is included in the full 5-task campaign as a ceiling check - all three models median 1.0 on L1, path recall, and composite across three runs, so it no longer separates frontier performers. Headline rank therefore uses PEP, AMZN, NFLX, and KO, where citation quality (CITE_BROAD, CITE_HALLUC) and task-specific traps still produce meaningful spread (Claude 0.994 vs GPT-4o 0.896 in this pilot).

6. WIP / next steps

Task #6 via scaffold_task.py - publish gate now includes task schema coherence (P3-37).
Scale to 15-task MVD (5 companies x 3 task types) with expert sign-off pipeline.
Tighten L3 citation rules or eval-mode prompts - CITE_BROAD is the main separator on headline tasks.

Synthetic L3 at 3x runs / harder decoys before treating it as a ranking dimension.

Track B: more frontier trajectories + PM branch coverage.

LATER-06: full corpus ingest with excerpt SHA pins (P3-10 mitigation on KO bundle).

References: docs/ARCHITECTURE.md

benchmark_v0.1/docs/PILOT_EVAL_JUL2026.md

env_v1/docs/METHODOLOGY_RL_ENV.md